Chengtao Xu

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EDUCATION

Yale University *New Haven, CT, United States*
Postdoctoral Fellow, Molecular Biophysics and Biochemistry *Jun.2024-Present*

Cornell University *Ithaca, NY, United States*
Postdoctoral Associate, Molecular Biology and Genetics *Mar. 2023-Jun.2024*

Southeast University *Nanjing, China*
Ph.D., Biomedical Engineering *Sep. 2018- Mar. 2023*

Southeast University *Nanjing, China*
B.S., Biomedical Engineering *Sep. 2014-Sep. 2018*

RESEARCH EXPERIENCE

Yale University *New Haven, CT, United States*
Cornell University *Ithaca, NY, United States*
Postdoctoral Associate, Supervisor: Dr. Ailong Ke *Mar. 2023-Present*

- Converting IscB/Cas9 to a versatile RNA-guided RNA editor
- Structural insights to improve ssDNA-/RNA-targeting IscB/Cas9 in vivo performance
- Structural studies on CRISPR IVB/IVC complexes
- Structural studies on CRISPR I-E complexes for phage defense

Southeast University *Nanjing, China*
Graduate Researcher; Advisor: Prof. Hong Liu *Jun. 2019- Mar. 2023*
Versatile DNA-based data storage based on the principle of movable type printing
Integrated DNA data storage system based on electrochemical DNA synthesis and sequencing

Southeast University *Nanjing, China*
Graduate Research Assistant; Advisor: Dr. Biao Ma *Sep. 2018- Jun. 2019*
High-resolution patterning of liquid metal on hydrogels to construct flexible electronics

Southeast University *Nanjing, China*
Undergraduate Research Program; Advisor: Prof. Hong Liu *Sep. 2017-Sep.2018*
Ultrasensitive arsenic detection based on catalytic reduction of Rhodamine B

PEER-REVIEWED PUBLICATIONS

Chengtao Xu, Xiaolin Niu, Haifeng Sun, Hao Yan, Weixin Tang, Ailong Ke* Conversion of IscB and Cas9 into RNA-guided RNA editors. *Cell*, 2025, 188, 21, 5847-5861.

Chengtao Xu, Qi Yang, Xiaolin Niu, Ailong Ke* Structure basis for single-strand nucleic acid targeting by IscB and variants. Under submission.

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Conor Pittman#, **Chengtao Xu#**, Ryan J. Catchpole, Sandra Garrett, Ryan Fuchs, Xiaomeng Chu, Kira S. Makarova, Eugene V. Koonin, Peng Zhao, Lance Wells, Brenton R. Graveley, Ailong Ke*, Michael P. Terns* Type IV-C CRISPR-Cas effector complexes recognize double-stranded DNA and switch on collateral cleavage of ssDNA and RNA. Under submission.

Chengtao Xu, Biao Ma, Zhongli Gao, Xing Dong, Chao Zhao, Hong Liu* Electrochemical DNA synthesis and sequencing on a single electrode with scalability for integrated data storage. *Science Advances*, 2021, 7, eabk0100.

Chengtao Xu†, Chao Zhao†, Biao Ma and Hong Liu* Uncertainties in synthetic DNA-based data storage. *Nucleic Acids Res.*, 2021, 49, 10, 5451-5469. (Review)

Chengtao Xu, Biao Ma, Xing Dong, Lanjie Lei, Qing Hao, Chao Zhao,* Hong Liu,* Assembly of Reusable DNA Blocks for Data Storage Using the Principle of Movable Type Printing. *ACS Appl. Mater. Interfaces*, 20230, 15, 20, 24097-24108.

Chengtao Xu†, Biao Ma†, Shuai Yuan, Chao Zhao,* and Hong Liu* High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. *Adv. Electron. Mater.*, 2020, 6, 1900721.

Dezhi Feng, Biao Ma, **Chengtao Xu**, Jingyi Xu, Yiyang Zhang, Yi Chen, Gangsheng Chen, Yakun Gao, Duxin Chen, Wenwu Yu, Chao Zhao*, and Hong Liu*, Renewable electrode array based on transformable metal for scaling up DNA synthesis and manipulation. *Adv. Funct. Mater.*, 2025, 36, e13703

Yakun Gao, Biao Ma*, Gangsheng Chen, **Chengtao Xu**, Ziyan Kong, Yanjie Chen, Chao Zhao, Duxin Chen, Wenwu Yu, and Hong Liu*, Transformable and stimuli-responsive liquid metal for integrated, sustainable, and biomimetic DNA-based data storage. *Matter*, 2025, 8, 102145.

Biao Ma, **Chengtao Xu**, Lishan Cui, Chao Zhao* and Hong Liu* Magnetic Printing of Liquid Metal for Perceptive Soft Actuators with Embodied Intelligence. *ACS Appl. Mater. Inter.*, 2021, 13, 4, 5574-5582.

Biao Ma, **Chengtao Xu**, Junjie Chi, Jian Chen, Chao Zhao* and Hong Liu,* A Versatile Approach for Direct Patterning of Liquid Metal Using Magnetic Field. *Adv. Funct. Mat.*, 2019, 29, 1901370.

H-index: 12 (as of March 16 2026)

(Other publications please see: https://scholar.google.com/citations?user=6VnVUPcAAAAJ&hl=en)

GRANTS & AWARDS

Tom and Joan Steitz RNA Fellowship award Apr. 2026
Tom and Joan Steitz RNA Fellows program 2026 class, mentor: Ailong Ke, Yale University

Poster Award *Oct. 2025*
2025 Annual Retreat of the Yale Center for RNA Science and Medicine

National Scholarship *Nov. 2021*
Highest scholarship given by Chinese government (Top 0.2%)

Excellent Bachelor Thesis Award *Sep. 2018*
Southeast University and Jiangsu province (Top 1%)

Presidential Scholarship *Dec. 2017/Nov. 2015*
Southeast University (Top 1%)

Chien-shiung Wu Scholarship for Biomedical Engineering *Jun. 2015*
Southeast University (Top 2%)

First Prize, National Innovation and Design Competition of Biomedical Engineering *Nov. 2016*
Designed a portable detection platform based on a microfluidic paper chip (Top 3%)

POSTERS & PRESENTATIONS

Chengtao Xu and Ailong Ke* Conversion of IscB and Cas9 into RNA-guided RNA-editors. Poster award. 2025 Annual Retreat of the Yale Center for RNA Science and Medicine. New Haven, United States, November 2025.

Chengtao Xu and Ailong Ke* Conversion of IscB and Cas9 into RNA-guided RNA-editors. Poster. Cold Spring Habour Laboratory Genome Engineering: CRISPR Frontiers. New York, United States, August 2025/2026.

Chengtao Xu and Hong Liu* Integrated data storage system based on DNA electrochemical synthesis and sequencing. Oral presentation. Joint Forum on Bioelectronics and Biophotonics. Hainan, China, December 2021.

Chengtao Xu†, Biao Ma†, and Hong Liu* High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. Online oral presentation. International Conference on Biomedical and Bioinformatics Engineering. N0028. Kyoto, Japan, November 2020.

Chengtao Xu†, Biao Ma†, and Hong Liu* High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. Poster. 3rd International Conference on Flexible Electronics. 103. Hangzhou, China, July 2019.

Chengtao Xu and Hong Liu Ultrasensitive Point-of-Care Testing of Arsenic Based on Catalytic Reaction of Unmodified Gold Nanoparticles. Poster. Asian Conference on Nanoscience and Nanotechnology. Qingdao, China, October 2018.

LEADERSHIP & COMMUNITY SERVICE

Mentoring, Ailong Ke Group, Yale University / Cornell University *Mar. 2023-Present*

- Mentored rotation graduate students in CRISPR biology and protein engineering, fostering scientific enthusiasm and rigorous research standards
- Mentored visiting undergraduate students at Cornell during summer research programs, supporting independent project completion

Scientific Community Engagement, Yale University *Jun. 2024-Present*

- Presented research in the Yale MB&B Research-in-Progress seminar series and RNA Club.
- Organized departmental Friday scientific communication sessions to facilitate cross-group discussion.
- Participated in Yale RNA Center annual retreat.

Mentoring, Hong Liu Group, Southeast University, *Sep. 2018-Mar. 2023*

- Mentored undergraduate students through thesis projects, from experimental design to final writing.
- Introduced two incoming PhD students to independent research

Laboratory Assistant, Hong Liu Group, Southeast University *Dec. 2020-Present*

- Managed laboratory functions including organization, ordering, laboratory safety management, and

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scheduling equipment use.

- Trained new graduate students with basic experimental techniques including electrochemical measurements, gel electrophoresis, microfluidic device fabrication.
- Guided students on new research topics (e.g., using primer exchange reaction to amplify DNA aptamers for pathogen detection).

Minister of Sports Department, Graduate Student Union of School of Biomedical Engineering, Southeast University *Sep. 2019-Jun. 2020*

- Managed Sports Department functions including competition organization, ordering, staff coordination.

Minister of Study Department, Undergraduate Student Union of School of Biomedical Engineering, Southeast University *Sep. 2015-Jun. 2016*

- Managed Study Department functions including lesson guidance, providing study materials, staff coordination.